

èfjç» - ç” µè`❖i¼š188-5665-3017 - é,❖±i¼šhs3434@foxmail.com -
ä, ºäºç½ ‘ç«™i¼šhs3434.github.io - **æ±
,è❖Ææ,,❖ä❖ “**i¼šAMD ROCm Python ä❖Žç«`ä¼ä❖ ‘ä • ¥ç`<ä, ^i¼^ROCm Radeon Cloud
æ-¹ä❖ ‘i¼% --- ## ä, ºäºç❖€ä» 3 ä¹´ç” Ýç%©äziæ❖`ä • ¥ç`<äÆ-ç»❖éäEi¼Æè ± ää|è½-äž<
K8s ä❖Žç«`æ ^ä€, \*\*äº¹ä™`äÆ-éf`ç½²ä❖èµ,,æº❖èºfäº|ä❖ä»»äŠiç¼-æž’ ä❖GPU
æž`ç❖ + ææ❖äŠi\*\*ä€, ææèèè; ‘2 ä ‘`è ± ä| ROCm / K8s / Helm / FastAPI /
Prometheusi¼Æç<-ç«<äºÆæ^❖ [**k8s-llm-runtime**](https://github.com/hs3434/k8s-llm-
runtime)i¼^K8s ä, Šçš,, vLLM æ`jāž<ææ❖äŠiç½ ‘ä...³i¼%oi¼Æ**v0.1.0
taggedi¼Æä❖`çžºäæºè • ‘demo**ä€, ç + Ýæ,% Snakemake ç±» Argo Workflows çš,, ML
ä¹³ä❖æš½èè±jä€, ## æšæææ-ä❖šä❖ - [Transformer ä»žä...¥é—`ä^ºä, Šæ%«](https://
hs3434.github.io/2025/03/03/transformer1/) - [GPT éžžæ ¼ Transformer è Š £ç ❖
EEGi¼šä, ºä»ä¹^ä❖ææžä¹^ä❖šäèèè, ©äº + ä»ä¹^ä❖ ‘](https://hs3434.github.io/
2025/06/08/transformer-eeg-decoding/) - [ä»žžé, ¶ æ❖ä, €ä¥—ç«`äºç«` EEG äziä❖ • ä»
,ç❖ + ä • ¥ä... • é “¾](https://hs3434.github.io/2025/06/08/eeg-signal-processing-toolchain/)
--- ## æ±,è❖Ææšèèf½i¼^äÆ%ä²—ä½❖äÆ¹é...❖æž’ äº❖i¼% | ç±»ä^« | æšèèf½ |
|-----|-----| | **ä❖Žç«`æ , äzi\*\* | Python 3.11+ä❖FastAPIä❖pytest | | \*\*äº¹ä™`ä , žç¼-
æž’ ** | Dockerä❖Kubernetesä❖Helmä❖é • æäfä» “äº “çºiç❖ + | | **GPU / LLM
ç” Ýæ❖** | vLLMä❖PyTorchä❖Hugging Face | | **äº`è Š , æµææ Š ** |
Prometheusä❖structlog | | **ä • ¥ä... • é “¾** | uvä❖mypyä❖GitHub Actions | | **èèç»`**
| Linux èèç»`´ | --- ## éi¹çºç»»éäÆ ### k8s-llm-runtimei¼šK8s ä, Šçš,, vLLM
æ`jāž<ææ❖äŠiç½ ‘ä...³ - è ± ää|demoéi¹çº» **é—éç~**i¼^ä-¹äº” JD è❖Æè´£ 1 +
3i¼%oi¼švLLM æ`jāž<éf`ç½²éæè GPU + äº¹ä™` + è • `ç” ±i¼Æä»æé~Ýäºšäºä...±äºæ—
¶ æ`jāž<çºiç❖ + äºæ❖æ❖, i¼ç” çº ¶ /äºæ»æé~Ýæf³ç” ” OpenAI ä...¼äº¹ API æ— ¶ é—
”æ Š »é«~ä€, i¼^çº´ æžžä-¹äº” JDi¼š”äº æ`jāž<èºç»f/æž`ç❖ +
Workshopä❖æ • ™ç”`ä❖äšÝèf½æ¼” çººi¼% **æž ¶ æž,,**i¼^ä❖Æ chart + ä , %ä±, lib +
é • æäfä¼¼^äÆ-i¼%oi¼š - `charts/llm-inference/`i¼švLLM è´Ýè½½ charti¼Æ**GPU vendor
ä` ± æ❖ä»...é❖ `values.yaml` **i¼^amd.com/gpu / nvidia.com/gpu / none
ä , %ææ❖i¼Ævalues æ~¾ä¼ä❖éè%æçºi¼% - `charts/llm-router/`i¼šFastAPI ç½ ‘ä...³ chart -
ConfigMap æ³”ä...¥ initContainer äš è½½²ä...ä¼Ý chart - RBAC ææèäºæ❖fé™i¼^Pod/
Service/ConfigMap read+write + Leasei¼% - ServiceMonitor äº`éè%ä❖`ç” ”i¼^Service
0.0.0.0:9090/metricsi¼% - `src/k8s\_llm\_runtime/`i¼š3 ä±, lib æš½èè±j -
`K8sJobOperator`i¼^ä½žžä±, i¼Æééšç” ” Job èºfäº|i¼% - `VLLMInferenceOperator`i¼^ä ,
ä±, i¼ÆHelm CLI subprocess + GPU vendori¼% - `ModelOperator`i¼^é«~ä±, i¼ÆOpenAI ä...
¼äº¹ + è ± ääš`éf`ç½²i¼% **4 äº æšæææ-äºç,¹**i¼š 1. **lib ä , %ä±, ä^ + ä±, æš½èè±
j**i¼^ä-¹äº” JD ä❖Žç«`æž ¶ æž,,èf½äši¼% - ä, Šä±, ModelOperator ä❖ä...³äzi
load_model(alias) / unload_model(alias) - ä, ä±, VLLMInferenceOperator äºèè... helm
ç» + èš, - ä½žžä±, K8sJobOperator ééšç” ” K8s Job èºfäº| 2. **Helm ConfigMap æ³”ä...¥
aliases**i¼^ä-¹äº” JD äº¹ä™`äÆ- + èµ,,æº❖èºfäº|i¼% - é❖ziä...❖æ-❖æ-jæ-ºäçžæ`jāž<éf½
rebuild é • æäfä¼¼^GPU é • æäfä 15-20GB rebuild 5-10mini¼% - ConfigMap
çš’ çº Š ç” Ýæ • ^i¼Æé ¶ é • æäfäè ± äºº 3. **K8s Lease ä^ + ä, fä¼ä❖é” ❖**i¼^ä-¹äº”
JD é + ç¾æèèç»`´ + HÄi¼% - coordination.k8s.io/v1 Lease é²æç Router
äºšä%ºææ-ä¹ ¶ ä❖ ‘éf`ç½²ä❖Æä, €æ`jāž< - é” ❖ç²’ äº|æÆ%º model
aliasi¼Æè ± ääš`èè ± ææÝ + ç»çº| 4. **ä , šäŠiçº Š Prometheus metrics**i¼^ä-¹äº” JD
ä¹³ä❖ä❖`è Š , æµææ Š i¼% - `chat_completions_duration_seconds`i¼^histogrami¼Æä❖«
status code æ ± ç¾äi¼% - `http\_requests\_total`i¼^counteri¼ÆæÆ%º method/route/statusi¼%º
- `models\_loaded`i¼^gaugei¼Æä❖æ~ in-memory statei¼% - ServiceMonitor äº`éè%èèè>
Grafana **GPU vendor ä^ ± æ❖ä demo**i¼^ä-¹äº” JD äš ä^ + GPU é + ç¾äi¼%oi¼š -

values.yaml` gpu.vendor=amd|nvidia|none` - AMDi¼šamd.com/gpu èµ„æ°❖è°få°|i¼ŒEROcM
vLLM é • œåf❖ - NVIDIAi¼šnvidia.com/gpu èµ„æ°❖è°få°|i¼ŒCUDA vLLM é • œåf❖ -
nonei¼šCPU only æ¨jå¼❖i¼^kind cluster é»è°¤i¼Œæœ-åœå¼€å❖ ‘ç’¨i¼%¨**è
¨¨é±❖ä¿❖è-❖**i¼š - 31 commits / 90 tests pass / coverage 88% - ruff + mypy strict + helm-lint
å...¨ clean - 2 Helm chartsi¼^llm-inference + llm-routeri¼%¨**è , ©å❖ ‘ä , Žä¿å¤❖**i¼š - kind
èš,ç,¹å¼±å + ...é • œåf❖åš é€Yi¼šcontainerdConfigPatches + DaoCloud mirror ### åY°ä°Ž
Snakemake çš,, ML å¹³å❖å • ¥ä¼œæµ❖æj + æž ¶i¼^ç” Yä¿jèfŒæ™-i¼%¨ -
èfŒæ™-i¼šSnakemake rule è-æ³ • æ— æ³ • å°žçŽ°æ¨jå❖—és❖æ,,❖ç,,å❖^ -
è°¾è°¾i¼šæ¨ ‘å¼f rule è-æ³ • i¼ŒŒç, ‘æž¥è°få + ...éf¨æž¥å❖£åš¨æ€❖ç” Yæ^❖
rulei¼Œèš ,èŒfè¾¼ “å...¥è¾¼ “å ± °ç±»åžå❖å❖-çšš - æœ¨å¼❖æ¼æž¥ - å...³è❖” i¼šJD
åšåå + éj¨æœ°å™¨åj¹ å¹³å❖ç” å❖ ‘ç»❖é°Œ“â€” â€” æœ-éj¹ç°æ¨ ML å¹³å❖æš¼è±
jçš,,å • ¥ç¨å°žè - µ ### è,, ‘æœ°æž¥å❖£ CNN èš £ç ❖å™¨i¼^PyTorchï¼%¨â€” â€” è ± åå -
|çš,,demoéj¹ç° - ç«-å^°ç«-æ— ¶ç©°å❖ • çšš - èš £ç ❖ EEG ä¿jå❖ • - PyQt å°žæ— ¶ GUI +
FastAPI æš¨éœ² - **ä» “å°¨ “**i¼š[github.com/hs3434/bci-pipeline-demo](https://github.com/
hs3434/bci-pipeline-demo) --- ## å • ¥ä¼œç»❖åž + ### ä ,šæµ • å°
¤é±Œå❖jå¿jæ❖-çšš ‘æšŒæœ°œ™❖å...-å❖ , i¼œç” Yç°œä¿jæ❖-å • ¥ç¨å , ^ i¼œ
2025.04 - è ±³ä»š - **Docker éf¨ç½²²**i¼šä ,šåšjå»£ç❖ Docker åŒ-i¼^åY°ä°Ž snakemake +
è ±³ç” æj + æž ¶i¼%¨ - **æœ°åšjç«-è¿❖ç»¨**i¼šLinux æœ°åšjå™¨æ—
¥å , ç»‘ æš¤i¼ŒNginx å❖❖å❖ ‘ä»£ç» + i¼Œsystemd æœ°åšjç°jç» +
å°¨ ‘å¹³å❖å • ¥å... • å¼€å❖ “i¼šä ,šåšjè ± åš¨åŒ-æµ❖ç¨+ç°—æ³ • å • ¥å... • ###
ä ,šæµ • æ-š æ¨ “ç” Yç°œäŒæ|çšš ‘æšŒæœ°œ™❖å...-å❖ , i¼œ
ç” Yç°œä¿jæ❖-ç” å❖ ‘å • ¥ç¨å , ^ i¼œ 2022.08 - 2024.07 - **å°¨ ‘å¹³å❖ç” Yä¿jå • ¥å...
• å¼€å❖ “**i¼šR / Python / Linux / Docker / å°¨ ‘äžYç” Y - **ä ,šåšjè ± åš¨åŒ-**i¼šåŒæ»å -
|è½-å½ • ç»,,ä ,šåšjæµ❖ç¨è ± åš¨åŒ-i¼Œæ¨jå❖—åŒ-ä€å❖-å¤❖ç”¨ -
å¹³å❖åœå°❖æ°❖i¼š[cloud.oebiotech.com](https://cloud.oebiotech.com/#/home) ###
ä ,šæµ • æ-š æ¨ “ç” Yç°œäŒæ|æœ°œ™❖å...-å❖ , i¼œç” Yç°œä¿jæ❖-å°žå¹ç” Y i¼œ
2021.10 - 2022.01 - å❖ • ç» + èfžè½-å½ • ç»,,i¼šè
¨¨æžšš æ€❖å , èš ,å¨ + æžæ€æ❖” °åžžæ€ä , æ€šš åŒ-å¨ + æž❖ -
ä ,»è❖æšŒæœ-i¼šå❖ • ç» + èfžè½-å½ • ç»,,æ€Ræ€Pythonæ€Linux --- ##
æ • ¢è,²èfŒæ™- **è¥¿jåŒ—å + œæž-çšš ‘æšŒåššš åj** i¼œ æ¤ç°œçšš ‘å -
|ä ,žæšŒæœ-ä , “ä ,š i¼œ æœ-çšš ‘ i¼œ 2018.09 - 2022.07 i¼œ GPAi¼š3.35 - æ¨ • ä ,šè°æ-
± i¼ššæššåY°ä°Žé€è¾¼ ‘æ-è’ ,åžžå½ çj°åšå°❖é°|å€” ä¼❖ç , å...³æ€ššçš ¶ åššæšš—
å€” æ€ššçš ¶ æŒ ± æ ± æ« - åšš å¨éj¹ç°i¼šæ¹ç°èŒŒæž¥çšš ❖å¹ç ‘èš±èœèè “¿æšš—
é “œæ±jæY “çš,,å½±å “❖i¼^å❖,ä ,žå½œèè...i¼%¨ --- ## ä ,å°°å¼~åš¿ - **2 å¨¨è ± å| K8s
å❖žç«-æ¨**i¼šä»žé¶¶ å¨ v0.1.0 tagged éj¹ç°i¼ŒŒç«-ç««å°Œæ^å❖Œ Helm chart + Python
lib - **çž°åœ° demo èf½åš¨**i¼šå❖-è ± åå ,|ç” µè,, ‘åœ¨¨ kind cluster ä ,šè • ‘éššå°Œæ • ‘
LLM æž¨ç» + æµ❖ç¨i¼^éf¨ç½²² å + ’ è - ç” ± å + ’ chat completions å + ’ metricsi¼%¨ -
åjå¹ èf½åš¨å¼°i¼šè ± åå|ç” Yç°œä¿jæ❖-æ€❖ç°—æ³ • æ€K8s ç -
°œåššéŒ + åYŸçY¥è- + i¼ŒŒç«-ç««å°Œæ^éj¹ç° - **3 å¹ Python å • ¥ç¨åŒ-
ç»❖é°Œæ°❖i¼šä»žçšš ‘ç” è,,šæœ-å¨°ä° ‘å¹³å❖å • ¥å... • çš,,å°Œæ • ‘è - å¾¼,,